Entity-Relationship Modelling

1. Entity and Entity Set

An entity is a real-world object or thing that can be clearly identified and about which the database stores data, such as a student, a course or a lecturer. An entity set is the collection of all entities of the same type.

2. Attributes

Attributes are the properties that describe an entity. A composite attribute can be divided into smaller parts. A multivalued attribute can hold more than one value for a single entity. A derived attribute does not need to be stored because it can be worked out from another attribute.

3. Relationship

A relationship is an association between two or more entities, for example a student enrols in a course.

4. Cardinality

Cardinality describes how many entities on one side can be associated with entities on the other side. In a one-to-one relationship, each entity is linked to at most one entity on the other side, for example a person and a passport. In a one-to-many relationship, one entity is linked to many entities on the other side, for example one lecturer teaches many courses. In a many-to-many relationship, entities on both sides can be linked to many, for example students and courses.

5. Primary Key versus Foreign Key

A primary key is an attribute that uniquely identifies each entity. A foreign key is an attribute in one table that refers to the primary key of another table, and it is how a relationship is shown when the model is turned into tables.

6. Weak Entity

A weak entity does not have a key of its own that is enough to identify it, so it depends on another entity. An example is a dependent of an employee.